SECTION-6

CABLES

1.0 SCOPE: Refer Clause-1 of Section-I, General

1.1 This specification provides for the design, manufacture, testing, inspection and testing before despatch, supply and delivery, Free at Destination Stores of unarmoured, multi core, PVC insulated and sheathed copper control cable for use on AC single phase or three phase (earthed or unearthed) systems of rated voltages up to and including 1100V or DC systems of rated voltages up to and including 1500 V to earth specified herein for their satisfactory operation in different substations in Andhra Pradesh in India.

Sizes of the LT copper control cable are

i) 2c x 2.5 sq.mm
ii) 4c x 2.5 sq.mm
iii) 6c x 2.5 sq.mm
iv) 10c x 2.5 sq.mm
v) 12c x 2.5 sq.mm

Sizes of LT Aluminium power cables are
i) 3.5 Core 50 Sq.mm Al power cable
ii) 3.5 Core 900 Sq.mm Al power cable
iii) 3.5 Core 120 Sq.mm Al power cable

1.2 The Control Cable shall conform in all respects to high standard of Engineering design ,workmanship and latest revision of relevant standards at the time of offer and the purchaser shall have the Control to reject any work or material, which in his judgment is not in full accordance there with.

2.0 STANDARDS:

The control cable shall conform to the latest revisions with amendments with available of relevant standards, rules and codes some of which are listed hereunder for ready reference, unless otherwise specified else where in this specification.

IEC : 228	:	Conductors of insulated cables
IS:1554 (Part-1)	:	Specification for PVC insulated (heavy duty) electric cables for working voltages upto and including 1100 volts.
IS:5831	:	Specification for PVC insulation and sheath of electric cables.
IS:8130	:	Specification for conductors for insulated electric cables and flexible cords.
IS:10810	:	Specification for methods of test for Cables.
IS:3961 (Part-V)	:	Recommended current ratings for cables (PVC insulated light duty cables)
IS:3975	:	Specification for Galvanized Steel wires, Formed wires and Tapes for armoring of cables
IS:10418	:	Specification for Drums for Electric cables
IS:613	:	Specification for Copper Rods and Bars for

Electrical Purposes

Unless otherwise specified elsewhere in this specification, the control cables shall conform to the latest revisions with amendments with available of relevant standards, rules and codes some
of which are listed hereunder for ready reference, unless otherwise specified elsewhere in this specification.

3.0 CLIMATIC CONDITIONS: Refer Clause-2 of Section-I, General

4.0 MAIN FEATURES:

4.1 These general purpose insulation cables shall be suitable for use where the combination of ambient temperature and temperature rise due to load results in conductor temperature not exceeding

-- 70 deg. C for normal continuous operation
-- 160 deg. C for short circuit condition

4.2.1 The cables shall be suitable for laying in covered/open trenches, ducts, trays or buried directly underground under dry or wet conditions. Suitable additives to
prevent attack by rodents, termites shall be added to the outer sheath PVC compound. Also, the outer sheathing shall be so designed as to afford high degree of mechanical protection besides making it resistant to heat, oils, chemicals, abrasion, weather conditions etc. Common acids, alkalis, contaminants saline solutions etc., shall not have adverse effects on the PVC sheathing material used.

The cables shall be capable of operating continuously under the system frequency variation of + or - 5% voltage variation of + or - 10% and a combined frequency - voltage variation of + or - 10%.

5.0 TECHNICAL REQUIREMENTS

5.1 CONDUCTOR

5.1.1 MATERIAL:

The conductor shall be composed of plain annealed copper wires(complying with IS:8130) made of high conductivity copper rods(complying with the latest version of IS:613).

5.1.2 FORM:

The conductors shall be circular, stranded. They shall be clean, reasonably uniform in size and shape, smooth and free from all harmful defects.

5.1.2.1 JOINTS:

Joints shall be permitted in the individual wires of which the conductor is formed but no joint shall be within 300mm of any other joint within the same layer. The joints shall be made by resistance butt welding ,fusion welding, cold pressure welding, electric welding, gas welding, brazing or silver soldering.

5.1.2.2 Classification:

Class-2 for stranded circular non-compacted cables used for fixed installation.

5.1.3 SIZE, NUMBER, RESISTANCE:

The wires in the conductor shall have the same nominal diameter after stranding(0.67mm). The number of wires in the conductor shall be not less than 7 (Seven). The minimum diameter of each wire after stranding at any point, through-out the length of the cable shall not be less than the nominal diameter by 0.01 mm.and the maximum resistance of conductor at 20 deg. C shall be 7.41 Ohms/km as per Table 2 of IS:8130.Nominal diameter of each strand after stranding shall be 0.68mm.

5.2 INSULATION:

5.2.1 MATERIAL:

The insulation shall be "General Purpose Insulation" of Type-A PVC Compound (conforming to the requirements of IS-5831) suitable for a maximum rated conductor temperature of 70 deg C intended for cables with rated voltages up to and including 3.3 kV. The conductor shall be provided with this Type-A PVC insulation applied by extrusion.

5.2.2 COMPOSITION:

The insulation shall consist of a compound based on one of the following materials, which have been SUITABLY COMPOUNDED AND PROCESSED to meet the requirement of the IS:5831

a) Polyvinyl chloride (PVC)

b) Suitable co-polymers of which the major constituent shall be vinyl chloride.

c) Mixture of (a) and (b)

5.2.3 TEST REQUIREMENT:

The insulation shall satisfy the test requirements stated in columns 2,3,4, & 7 of TABLE-1 of IS:5831.

5.2.4 THICKNESS:

The average thickness of insulation shall be not less than the nominal value (t1) of 0.9 mm/1.2 mm applicable for un-armoured/armored multi-core cables of size 2.5 sq.mm as per Table 2 as specified in IS:1554 (Part I).

5.2.5 TOLERANCE ON THICKNESS:

The smallest of the measured values of thickness of insulation shall not fall below the

nominal value (t1 in. mm) by more than 0.1mm + 0.1 t1.

5.2.6 APPLICATION:

The insulation shall be so applied that it fits closely on the conductor and it shall be possible to remove it without damage to the conductor.

5.3 CORE:

5.3.1 IDENTIFICATION:

Cores shall be identified by different coloring of PVC insulation by adopting the following scheme:

a) 2 cores : Red and Black

b) 4 cores : Red, Yellow, Blue and Black

c) 6 cores/10/12 cores: Two adjacent core(counting and direction core) in each layer, blue and yellow remaining cores gray.By numbers in which case the insulation of all the cores shall be of same color and numbered sequentially, starting with number-1 for the inner layer. All the numbers shall be of the same color which shall contrast with the color of the insulation. The numerals shall be legible.

5.3.2 ARRANGEMENT OF MARKING:

The numbers shall be repeated at regular intervals along the core, consecutive numbers being inverted in relation to each other. When the number is a single numeral, a dash shall be placed underneath it. If the number consists of two numerals, these shall be disposed one below the other and a dash placed below the lower numeral. The spacing "d" between consecutive numbers shall not exceed 50mm as shown in fig. 1 of IS:1554 (Part-I)-1998 Page-4(core identification by numbers).

5.4. LAYING UP OF CORES:

In twin and multicore cables, the cores shall be laid up together with a suitable lay; the outermost layer shall have right hand lay and the successive layers shall be laid with opposite lay, where necessary the interstices shall be filled with non-hygroscopic material.

The lay-up of core shall be applicable as recommended in Table-3 of the IS:1554 Part-1.

5.5 INNER SHEATH:

5.5.1 MATERIAL:

Inner sheath shall be as follows:

a) Un vulcanized rubber or
b) Thermoplastic materials, or
c) Proofed tape (for inner sheath only)

Un vulcanized rubber or thermoplastic material used shall not be harder than the type-A PVC used for insulation and the Type ST- 1 PVC used for outer sheath. The material shall be so chosen as to be compatible with temperature ratings of the cable

and shall have no deleterious effect on any other component of the cable.

5.5.2 CONSTRUCTION:

The laid up cores shall be provided with inner sheath applied by extrusion only. It shall be ensured that the shape is as circular as possible. The inner sheath shall be so applied that it fits closely on the laid up cores and it shall be possible to remove it without damage to the insulation. The thickness of the inner sheath shall be as specified in column 3 of Table-4 of IS:1554, applicable for the multicore cables whose calculated diameter over laid up cores is upto and including 25mm and above. If and when one or more layers of binder tapes are applied over the laid up cores, the thickness of such tapes shall not be construed as part of the inner sheath.

5.6 OUTER SHEATH:

5.6.1 MATERIALS:

For cables with general purpose insulation, the outer sheath shall be of Type ST-1 PVC compound conforming to the requirements of IS:5831 as recommended for general purpose sheath intended for use in cables operating at a maximum rated conductor temperature 70 deg C. The test requirements shall be as per Columns 2,3,4 & 6 of Table-2 of IS:5831.

5.6.2 CONSTRUCTION:

This outer sheath shall be applied by extrusion over the inner sheath as applicable for unarmoured twin and multicore cables. It shall be so applied that it fits closely over inner sheath and it shall also be possible to remove it without damaging the inner sheath. The color of the outer sheath shall be black.

5.6.3 THICKNESS:

The average and minimum thickness of the PVC outer sheath shall be not less than the values specified in col. 3 , 4 & 5 of Table 7 of IS: 1554.

5.6.4 Armoured Cables: The armour shall be of galvanised steel wire and round wire armouring for control cables. The outer sheath shall have FRLS properties.

6.0 TESTS:

6.1. TYPE TESTS:

6.1.1 The cables offered should have been fully type tested as per IS/IEC (with latest amendments) in a recognized laboratory. The date of type tests conducted shall be the latest and not earlier than Ten years as on the date of bid opening. The bidder shall furnish one set of type test reports along with the offer.

6.2 ACCEPTANCE TESTS:

The following shall constitute acceptance tests

a) Annealing test
b) Conductor resistance
c) Thickness of insulation and sheath

d) Tensile strength and elongation at break of insulation and sheath
e) Insulation resistance test
f) High voltage test at room temperature
g) Flammability
h) Oxygen & Temp index test
i) Test for armouring wires

The sampling plan for acceptance tests shall be as per Appendix-A of the IS:1554 (Part-1)

6.3 ROUTINE TESTS :

The following shall constitute routine tests:

a) Conductor resistance test
b) High voltage test at room temperature

6.4 OPTIONAL TESTS:

a) Cold bend
b) Cold impact

NOTE: i) For test methods, relevant parts of IS - 10810 shall be referred.

ii) The test certificates as a result of the inspection of the cables shall be furnished to the purchaser for his approval prior to despatch.

7.0 INSPECTION

7.1 The Purchaser's representative shall, at all times, be entitled to have access to the works and all places of manufacture where cable shall be manufactured and the representative shall have full facilities for unrestricted inspection of the Bidder's works, raw materials and process of manufacture and conducting necessary tests as detailed herein.

7.2 The Bidder shall keep the Purchaser informed in advance of the time of starting and of the progress of manufacture of cable in its various stages so that arrangements can be made for inspection.

7.3 No material shall be despatched from its point of manufacture before it has been satisfactorily inspected and tested, or unless the inspection is waived by the Purchaser in writing. In the later case also, the cable shall be despatched only after satisfactory testing for all tests specified herein has been completed and approved by the Purchaser.

7.4 The acceptance of any quantity of material shall in no way relieve the Bidder of any of his responsibilities for meeting all requirements of the specification, and shall not prevent subsequent rejection if such material is later found to be defective.

7.5 At least 5% of the total number of drums subject to minimum of two in any lot put up for inspection, shall be selected at random to ascertain the length of cable by the following method:

"At the works of the manufacturer of the cable, the cable shall be transferred from one drum to another at the same time measuring its length with the help of a graduated pulley & Cyclometer. The difference in the average length thus obtained and as declared by the Bidder in the packing list shall be applied to all the drums if the cable is found short during checking."

7.6 The bidder shall give 15 days advance intimation to enable the purchaser to depute his representative for witnessing acceptance and routine tests. All the charges in connection with inspection by purchaser personnel such as travel, accommodation and incidentals shall be borne by the purchaser.

8.0 QUALITY ASSURANCE PLAN :

8.1 The bidder shall invariably furnish the following information along with his offer failing which the offer shall be liable for rejection. Information shall be separately given for individual type of equipment offered.

i) The Structure of organization
ii) The duties and responsibilities assigned to staff ensuring quality of work
iii) The system of purchasing, taking delivery and verification of materials
iv) The system for ensuring quality of workmanship
v) The quality assurance arrangements shall confirm to the relevant requirement of ISO-9001 on ISO 9002 as applicable.
vi) Statement giving list of important raw materials, names of sub-supplies for the raw materials, list of standards according to which the raw material are tested, list of tests normally carried out on raw material in the presence of suppliers representative, copies of test certificates.
vii) Information and copies of test certificates as on (i) above in respect of bought out items.
viii) List of manufacturing facilities available
ix) Level of automation achieved and list of areas where manual processing exists.
x) List of areas in manufacturing process, where stage inspections are normally carried out for quality control and details of such tests and inspections.
xi) The List of testing equipment available with the bidder for final testing of equipment specified and test plant limitation, if any vis-a-vis the type. Special acceptance and routine tests specified in the relevant standards. These limitations shall be very clearly brought out in "Schedule of Deviations" from the specified test requirements.

8.2 The Contractor shall within 30 days of placement of order, submit the following information to the purchaser.

i) List of the raw material as well as bought out accessories and the names of sub-suppliers selected from those furnished along with the offer.

ii) Type test certificates of the raw material and bought out accessories if required by the purchaser.

iii) Quality Assurance Plan (QAP) with hold points for purchasers inspection. QAP and purchasers hold points shall be discussed between the purchaser and contractor before the QAP is finalised.

The contractor shall submit the routine test certificates of bought out accessories and central excise asses for raw material at the time of routine testing if required by the purchaser and ensure that the quality assurance requirements of specification are followed by the sub-contractor.

8.3 The Quality Assurance Programme shall give a description of the Quality System and Quality Plans with the following details.

i) Quality System:

- The Structure of the Organisation
- The duties and responsibilities assigned to staff ensuring quality of work.
- The system of purchasing, taking delivery of verification of materials.
- The system of ensuring of quality workmanship.
- The system of control of documentation.
- The system of retention of records.
- The arrangement of contractor internal auditing.
- A list of Administrator and work procedures required to achieve contractors quality requirements. These procedures shall be made readily available to the purchaser for inspection on request.

ii) Quality Plans:

a. An outline of the proposed work and program sequence.
b. The structure of contractors organizations for the contract.
c. The duties and responsibilities ensuring quality of work
d. Hold and notification points.
e. Submission of engineering documents required by this specification.
e. The Inspection of the materials and components on request.
f. Reference to contractors work procedures appropriate to each activity.
g. Inspection during fabrication/construction.
h. Final inspection and test.

9.0 DOCUMENTATION:

9.1 All drawings shall conform to International Standards Organization (ISO) "A" series of drawing sheets/Indian Standards Specifications. All drawings shall be in ink and suitable for microfilming. All dimensions and data shall be in S.I. Units. **In addition to any other drawings which the tenderer may like to supply three sets of the followings drawings/manuals must be submitted with the tender.**

i) Drawing showing sectional view of cables and cable drums

ii) Technical and descriptive literature giving details of the cables offered

iii) Tables of rating factor for correction of current rating for variation in standard condition of installation.

iv) Test facilities available at the tenderer's works.

9.2 TEST REPORTS:

i) Four copies of acceptance test reports shall be furnished to the purchaser. One copy will be returned, duly certified by the purchaser and only thereafter shall the materials be despatched.

ii) All records of routine test reports shall be maintained by the Bidder at his works for periodic inspection by the purchaser.

iii) All test reports of tests conducted during manufacture shall be maintained by the Bidder. These shall be produced for verification as and when requested for by the purchaser.

10. IDENTIFICATION, PACKING AND MARKING:

10.1.1 MANUFACTURER'S IDENTIFICATION:

The manufacturer shall be identified throughout the length of the cable by means of a tape bearing the manufacturer's name or trade mark, by being indented, printed or embossed on the outer sheath of the cable

10.1.2 CABLE IDENTIFICATION:

The words "APTRANSCO" shall be embossed and printing of other details shall be done with Inkjet/lase printer throughout the length of the cable at frequent intervals (not more than 1 meter in any case) on the outer sheath.

10.1.3 CABLE CODE:

Cable code shall be as per IS: 1554 to indicate Copper cable having PVC insulation PVC outer sheath.

10.2.1 PACKING:

The cable shall be wound on a non-returnable wooden/steel drum (conforming to the latest version of IS:10810) of suitable size and packed. The ends of the cable shall be sealed by means of non-hygroscopic sealing material. The drum packing shall be robust enough for rough handling that may be subjected to during transit by water/rail /road. Type and material of the drum shall be specified by the Bidder by means of a dimensional drawing.

10.2.2 STANDARD DRUM LENGTHS AND QUANTITY TOLERANCE:

Each cable drum shall contain one SINGLE CONTINUOUS LENGTH. The desired drum lengths shall be as follows.

2Cx2.5 Sq.mm – 1000mts +/- 10%

4Cx2.5 Sq.mm – 1000mts +/- 10%
6Cx2.5 Sq.mm – 1000mts +/- 10%
10Cx2.5 Sq.mm – 1000/500mts +/- 10%
12Cx2.5 Sq.mm – 1000/500mts +/- 10%

Overall quantity tolerance shall be + or - 5%.
The Purchaser reserves the right to alter the drum lengths, if required, while placing the order. The bidder may also offer the maximum drum lengths he is capable of supplying against each size.

10.3 DRUM MARKING:

"THE CABLE DRUM SHALL BE DULY "IS" MARKED" and carry the following details:

a) Reference to the "IS"
b) Manufacturer's name, brand name, trade name
c) Type of cable and voltage grade
d) Number of cores
e) Nominal cross-sectional area of the conductor (Sq.mm).
f) Cable code.
g) Color of cores
h) Length of the cable on the drum (Mtrs.)
i) Direction of rotation of drum by means of arrow.
j) Approximate gross mass (Kg)
k) Running end of cable
l) Country of manufacturer and
m) Year of manufacture

11.0 PACKING AND FORWARDING: Refer Clause-4 of Section-I, General

12.0 QUANTITY AND DELIVERY REQUIREMENTS : Refer Clause-5 of Section-I, General

13 SUPERVISION SERVICES : Refer Clause-6 of Section-I, General

14. GUARANTEED TECHNICAL PARTICULARS

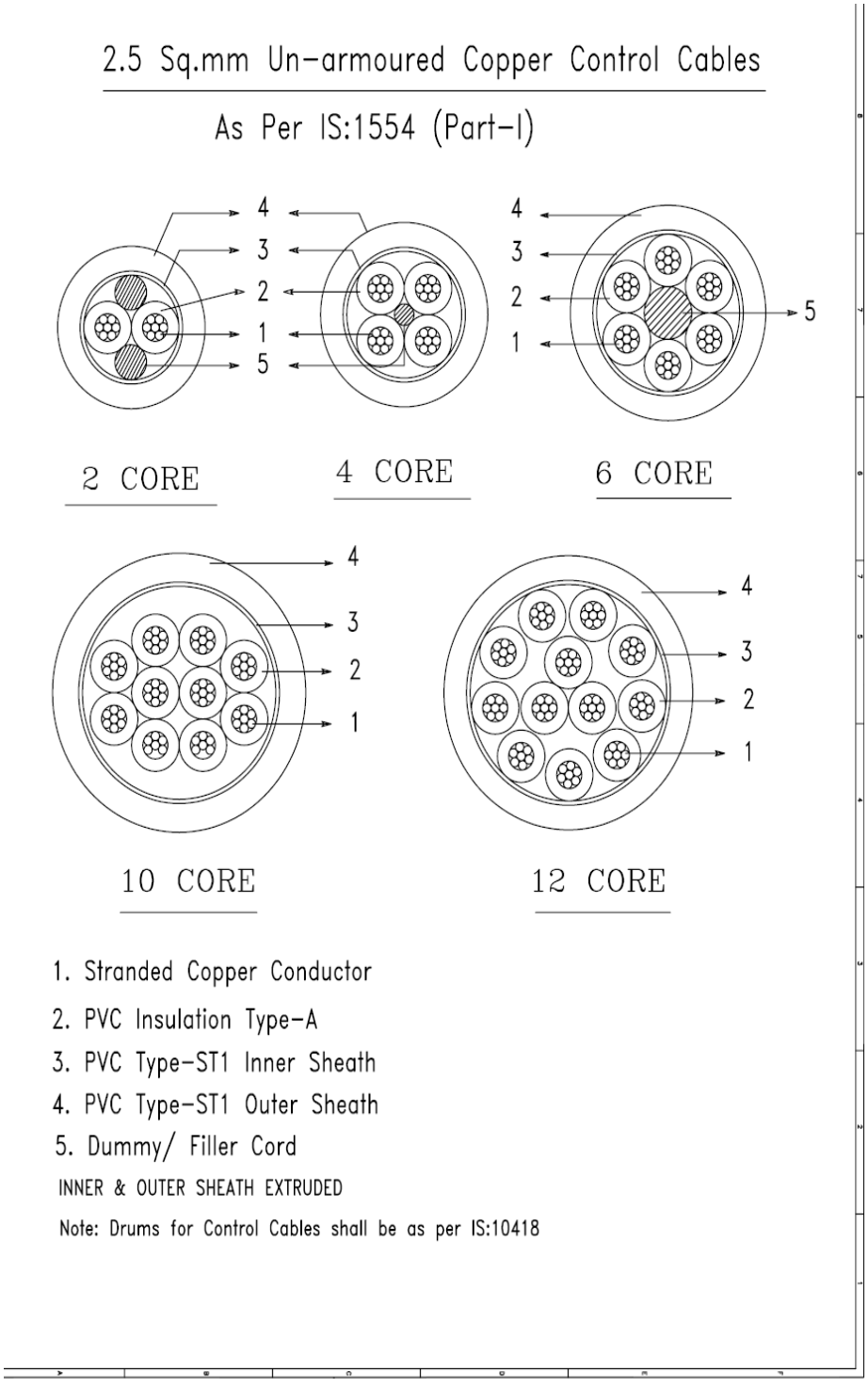
The bidders shall furnish the Guaranteed Technical particulars as per Annexure enclosed. The bids received without the filled in Guaranteed Technical Particulars shall be treated as non responsive and rejected.

ANNEXURE
GUARNTEED TECHNICAL PARTICULARS

1	Manufacturer's name	
2	a) Type and description of the cable with size	
	b)Standards which they conform to	
3	CONDUCTOR (Sq.mm)	2Cx2.5 4Cx2.5 6Cx2.5 10Cx2.5 12Cx2.5
	a) Material	
	b) Whether stranded	
	c)If so, number of strands	
	d)Diameter of each strand after stranding(mm)	0.67mm
	e)Max.resistance at 20 Deg.C(Ohms/Km)	
4	INSULATION	
	a) Material	
	b)Nominal thickness(mm)	
	c)Minimum tensile strength without ageing (N/mm2) and maximum % variation after ageing	
	d)Minimum elongation at break without ageing (%) and maximum % variation after ageing.	
	e)Minimum volume resistivity at	
	i) 27 DegC (Ohm-Cm)	
	ii)Max.rated temperature of 70 Deg.C(Ohm-Cm)	
	f)Minimum insulation resistance constant at	
	i) 27 Deg C(Mega ohm/Km)	
	ii)Max.rated temperature of 70 Deg.C	
	(Mega ohm/Km)	
	iii)Whether application of insulation is by	
	way of extrusion	
5	INNER SHEATH	
	a) Material	
	b)Minimum thickness inner sheath(in mm)	
	c)Whether method of application is by	
	way of extrusion	
6	Interstices(or fillers)	
	a)Whether used in the formation of cable	
	b)Material	
7	Binder tape(Over the laid up cores)	
	a)Material	
	b)Thickness(mm)	

8	OUTER SHEATH	
	a)Material	
	b)Nominal/Minimum thickness(mm)	
	c)Minimum tensile strength without ageing (N/mm2) and maximum % variation after ageing	
	d)Minimum elongation of break(%) and maximum variation after ageing (%)	
	e)Whether method of application is by	
	way of extrusion	
	f)Are the inner and outer sheaths extruded in a single operation out of the material intended for outer sheaths	
	g)Whether the PVC suitably treated for withstanding the working conditions.	
	h) Colour	
9	PHYSICAL PARAMETERS	
	a)Overall dia of core(mm)	
	b)Calculated diameter over laid up cores(mm)	
	c)Calculated diameter under the outer	
	sheath(mm)	
	d)Overall diameter of the finished cable(mm)	
	e)Drum length(mtrs)/tolerance (%)	
	f)Cable weight(Kg/Km)/Tolerance(%)	
	g)Gross weight of the complete drum	
	with cable(Kg)	
	h)Whether core/cable identification/marking as specified shall be carried out.	
10	Electrical parameters	
	a)Rated voltage(volts)	
	b) Voltage grade(volts)	
	c)Whether suitable for earthed/Unearthed system	
	d) Continuous current carrying capacity(Amps)	
	i) In Air	
	ii) In ducts	
	iii) In ground	
	e) Short circuit current capacity for 1 sec. Duration	
	f) Max. conductor temperature during short circuit condition.	
11	WEIGHTS & DIMENSIONS etc. : 2C 4C 6C 10C 12C 5C 14C a) Nominal length of cable per drum (Tolerance ±10%) mtrs. b) Cable weight (Tolerance : ± 5%) kg / km c) Gross weight of complete drum with cable (Aproximate) kg d)Whether core/cable identification/marking as	

	specified shall be carried out e)Whether method of application is by way of extrusion f)Are the inner and outer sheaths done in single operation and of the material intended for outer sheathes As per IS:1554 (Part-I) g)Whether sequential marking of length provided on the outer sheath	



A C R O N Y M S

MVA	Mega Volt Amperes
KVA	Kilo Volt Amperes
KV	Kilo Volt
Cu. M	Cubic meter
deg. C	Degrees Centigrade
mm.	Milli meter
cm.	Centi metre
dm.	Deci metre
km.	Kilo metre
g.	Grams
kg.	Kilo gram
V	Volts
Hz	Hertz
DC	Direct Current
AC	Alternate Current
ONAN	Oil Natural Air Natural
ONAF	Oil Natural Air Forced
HV	High Voltage
IV	Intermediary Voltage
Db	Decibel
KA	Kilo Amperes
Sec.	Seconds
Amps.	Amperes
kVp.	Kilo volts peak
kV. rms.	Kilo volts root mean square
KW	Kilo watt
Max.	Maximum
Min.	Minimum
UTS	Ultimate Tensile Strength
QAP	Quality Assurance Plan
ACSR	Aluminium Conductor Steel Reinforcement
AAAC	All Aluminium Alloy Conductor
HP	Horse power
ppm.	Parts Per Million

OLTC	On Load Tap Changer
WTI	Winding Temperature Indicator
OTI	Oil Temperature Indicator
Torr	Torr.
LOI	Letter of Intent
CIF	Cost including Insurance & Freight
CIP	Carrier Insurance Paid
SF6	Sulphur Hexa Flouride
TRV	Transient Recovery Voltage
CTD	Capacitor Tripping Device

ICU	Individual Compressor Unit
NO	Normally Open
NC	Normally Close
Sq.mm Al. Cable	Square millimeter Aluminium cable
Sq.mm Cu. Cable	Square millimeter Copper cable
MS Flat	Mild Steel Flat
C	Close
O	Open
HRC	High Rupturing Capacity
PVC	Poly Vinyl Chloride
p.f.	Power Factor
CT	Current Transformer
NCT	Neutural Current Transformer
TEFC	Totally Enclosed Fan Cooled
P.U.	Per Unit
ISO	Indian Standard Organisation
CB	Circuit Breaker
Auto	Automatic
kg./Sq.mm	Kilogram per square millimeter
T	Metric Tonn
KN	Kilo Newtons